

Software Engineering

BCA — Semester IV — 2024 — Answer Sheet

Subject Code: CACS253

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. Describe the challenges faced in software engineering.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain the principles and practices of Extreme Programming (XP) in software development and discuss its significance in modern software engineering methodologies.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is regression testing? How test cases can be designed? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is user interface design? Describe its main principles and the impact of a well-designed UI on user experience and software success.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Discuss the relationship between programming languages and development tools in software engineering, highlighting how they complement each other.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is Capability Maturity Model (CMM)? Describe levels of CMM.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Explain the concept of versioning in software engineering and discuss its importance in software development and maintenance.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Discuss Computer-Aided Software Engineering (CASE) tools, their types, features, and the benefits they bring to the software development process. Provide examples of popular CASE tools and explain how they are used in various stages of the software development lifecycle.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain the processes of requirements validation and requirements specification in the context of software development. Highlight their significance, methods used, challenges encountered, and provide examples to illustrate their application.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Discuss the concept of modular decomposition in software engineering. Explain its importance, the principles guiding effective modular decomposition, and provide examples to illustrate how modular decomposition can be applied in practice.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.